International Journal of Computer Assisted Radiology and Surgery (2024) 19:1259–1266
https://doi.org/10.1007/s11548-024-03171-6

ORIGINAL ARTICLE

OneSLAM to map them all: a generalized approach to SLAM for monocular endoscopic imaging based on tracking any point

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Received: 4 March 2024 / Accepted: 30 April 2024 / Published online: 22 May 2024
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Abstract

Purpose Monocular SLAM algorithms are the key enabling technology for image-based surgical navigation systems for endoscopic procedures. Due to the visual feature scarcity and unique lighting conditions encountered in endoscopy, classical SLAM approaches perform inconsistently. Many of the recent approaches to endoscopic SLAM rely on deep learning models. They show promising results when optimized on singular domains such as arthroscopy, sinus endoscopy, colonoscopy or laparoscopy, but are limited by an inability to generalize to different domains without retraining.

Methods To address this generality issue, we propose *OneSLAM* a monocular SLAM algorithm for surgical endoscopy that works out of the box for several endoscopic domains, including sinus endoscopy, colonoscopy, arthroscopy and laparoscopy. Our pipeline builds upon robust *tracking any point (TAP)* foundation models to reliably track sparse correspondences across multiple frames and runs local bundle adjustment to jointly optimize camera poses and a sparse 3D reconstruction of the anatomy.

Results We compare the performance of our method against three strong baselines previously proposed for monocular SLAM in endoscopy and general scenes. *OneSLAM* presents better or comparable performance over existing approaches targeted to that specific data in all four tested domains, generalizing across domains without the need for retraining.

Conclusion *OneSLAM* benefits from the convincing performance of TAP foundation models but generalizes to endoscopic sequences of different anatomies all while demonstrating better or comparable performance over domain-specific SLAM approaches. Future research on global loop closure will investigate how to reliably detect loops in endoscopic scenes to reduce accumulated drift and enhance long-term navigation capabilities.

Keywords Computer vision · Monocular SLAM · 3D motion estimation · Endoscopy · Arthroscopy · Image-based navigation · Tracking any point

Introduction

Endoscopy is a minimally invasive procedure that relies on natural orifices or small incisions to inspect the internal anatomy of a patient. During an endoscopic inspection, a small camera is inserted through natural body cavities or small incisions to observe the internal tissues [1]. Given the benefits for patients, including reduced morbidity and faster recovery times of these minimally invasive surgeries compared to open surgeries, the use of endoscopy has extended to different anatomies and surgeries, including laparoscopy for abdominal procedures [2], colonoscopy for colorectal polyps screening [3], arthroscopy for joint examination and treatment [4] and nasal endoscopes for functional endoscopic sinus surgeries for managing chronic sinusitis [5], among others. However, despite the benefits of endoscopic procedures, the constrained field of vision leads to reduced global context of the anatomy structure, making surgical navigation an essential component to reduce the risk of iatrogenic perforations and other complications during the surgery [6].

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Simultaneous Localization and Mapping (SLAM) addresses the problem of generating a geometric representation of an unknown environment given a stream of sensor inputs and localizing the sensor in that environment in real time. Considering the endoscopic use case, the sensor then translates to the endoscope itself while the information it supplies is high-resolution but monocular images of the scene. It has consistently been shown that current generalist state-of-the-art SLAM systems [7–10] offer an adequate level of performance on real-world monocular sequences but struggle when translating to medical scenarios [11, 12]. The major challenges that prevent generalist SLAM systems from performing well in endoscopy are repetitive yet feature-scarce textures of internal tissues combined with photometric inconsistency due to the light source moving with the endoscopic camera. To address these challenges, different methods for visual odometry [13] and 3D reconstruction [11, 14–16] have been proposed for endoscopy.

While these methods for camera tracking in endoscopy have demonstrated promising results, they come with considerable disadvantages that limit their applicability beyond the narrow scenario they were developed for. For example, ArthroSLAM and TaToo are not built for monocular endoscopy, requiring additional information such as robot kinematics, inertial measurements or stereo video. Moreover, many of these methods, such as SAGE-SLAM, EndoSLAM and TAToo, are based on deep learning approaches that require a large and diverse datasets for training in order to generalize to unseen data. Even if such dataset is available, the resulting model is optimized for a specific domain only, requiring retraining when translated to a different endoscopic domain, and limiting their direct application to a wide range of real-world medical scenarios.

Contributions Considering these limitations, we propose *OneSLAM* a SLAM pipeline that can generalize to different endoscopic domains without the need for retraining, all while performing better or comparably to domain-specific models across all tested domains. To achieve this objective, we leverage *tracking any point (TAP) foundation models*, to obtain sparse correspondences across multiple image frames. These correspondences are used to run local bundle adjustment to jointly optimize camera poses and anatomical geometry across the incoming endoscopic sequence. To the best of our knowledge, this pipeline is one of the first to utilize the *tracking any point (TAP)* task for SLAM, as well as the first monocular endoscopy SLAM pipeline to demonstrate out-of-the-box generalization capabilities to unseen domains.

To verify the generalizability of our method, we perform exhaustive evaluations on four different datasets representing three different endoscopic domains including the sinus, joints and colon. We compare the performance of our *OneSLAM* pipeline against ORB-SLAM3, SAGE-SLAM and EndoSLAM's EndoSFM-Learner in the four endoscopic domains and show that *OneSLAM* produces competitive results without retraining and using the same settings across all domains. The code of our pipeline is publicly available at https://github.com/arcadelab/OneSLAM.

Related work

Real-time tracking and reconstruction are essential components in computer-assisted surgical navigation as they allow surgeons to perform procedures with increased precision and safety. In this context, SLAM techniques are well matched to the surgical navigation domain by providing real-time localization, mapping and spatial awareness.

In medical applications, various SLAM algorithms are employed. ORB-SLAM3 [7], a prominent feature-based approach, is used in endoscopy, ureteroscopy and other surgical domains. However, it struggles with smooth surfaces like sinus, joints and colon, leading to failures. On the other hand, direct methods such as [8, 9] are feature agnostic and suitable for feature-scarce tissues. Nevertheless, they lack robustness in dynamic scenes and changing illumination conditions common in endoscopy. The semi-direct method SD-DefSLAM [17] employs photometric tracking and a deformable optimization backend for deformable medical sequences. However, it relies on a restrictive deformation assumption that may not always hold.

With the rise of deep learning, several learning-based methods were proposed for endoscopy. SAGE-SLAM [11] learns appearance and geometry priors to handle pairwise image alignment. Its design focuses on enhancing tracking robustness and achieving accurate dense reconstruction in monocular endoscopy. Endo-SFMLearner [15] leverages dense convolution structure and spatial attention mechanisms to estimate monocular depth and poses for endoscope. Deep EndoVO [18] presents a monocular visual odometry algorithm based on deep recurrent convolutional neural networks for endoscope localization. While promising in performance, the training requirement of these models is a time-intensive process and not suitable for immediate usage.

Keypoint tracking is vital in the context of SLAM, as it guides the localization and mapping procedure. Traditional point detectors [19–21] are commonly used in general vision tasks, but struggle with textureless regions, illumination changes and occlusions. While learning-based feature detectors [22] and descriptors [23, 24] have been shown to enhance repeatability and robustness in the face of drastic illumination changes present in endoscopy, they suffer from long training processes and generalization issues. CoTracker [25] is a state-of-the-art *tracking any point (TAP)* [26] foundation model, which allows pixel-level tracking in long video sequences. The versatility of the TAP task for point tracking has motivated us to harness the robust tracking capabilities

of TAP foundation models in the development of a SLAM pipeline (Fig. 1).

Methods

Problem formulation

Monocular SLAM describes the problem of generating a map of the observed scene and localizing the camera in real time, given only the images of a video stream V as input. To achieve this, *OneSLAM* employs two main processes: (a) a mapping process, responsible for optimizing a sparse point map $M \in \mathbb{R}^3$ and set of key frame poses, and (b) a tracking process, which is responsible for localizing the current frame with respect to M. In the following, $\boldsymbol{c}$ and $\boldsymbol{m}$ identify variables belonging to the localization and mapping process, respectively. The time of the current frame is denoted by T. Note that in our case, we assume the scene to be static and as such do not handle deformation or tool movement.

To initialize our SLAM algorithm, we first accumulate a set of $l_{\boldsymbol{m}}$ frames and run a mapping process iteration on them. For every subsequent frame $T > l_{\boldsymbol{m}}$, we first run the tracking process to localize (pose estimation) the current frame with respect to the sparse map. In addition, if the last mapping process occurred $l_{\boldsymbol{m}}$ frames in the past, we run the mapping process again.

Point tracking

We track sets of key points $K_t \subset \mathbb{R}^2$ for every frame I_t in the video sequence V using a TAP foundation model, which given a section $V' = (I_{t-l}, I_T)$ of length l from the video stream V and the set of key points K_{t-l} on frame I_{t-l} as an input estimate the key point locations for all frames $I_{t>t-l}$. We choose CoTracker as the specific TAP foundation model, as at the time of writing, it denotes the current state of the art [25] (view Online Resource 1—Appendix H for an evaluation of *OneSLAM* using TAPIR [27] as an alternative TAP foundation model). We use the weights provided in [25] with stride and window size set to four and eight, respectively, and provide Co-Tracker with a support grid of additional 2D points to further improve the tracking accuracy. To ensure consistent point tracks (i.e., each key point track uniquely identifies a 3D location in the anatomy) so we employ the visibility mask of Co-Tracker to terminate a key point track as soon visibility is lost.

Localization

To localize the current frame I_T relative to the map M, we track a maximum of $n_{\boldsymbol{c}}$ key points $\hat{K}_{T-1} \subset K_{T-1}$ from the frame I_{T-1} onto the frame I_T. Afterward, we employ a RANSAC-based PnP algorithm that optimizes the pose P_T by minimizing the total reprojection error (Eq. 1).

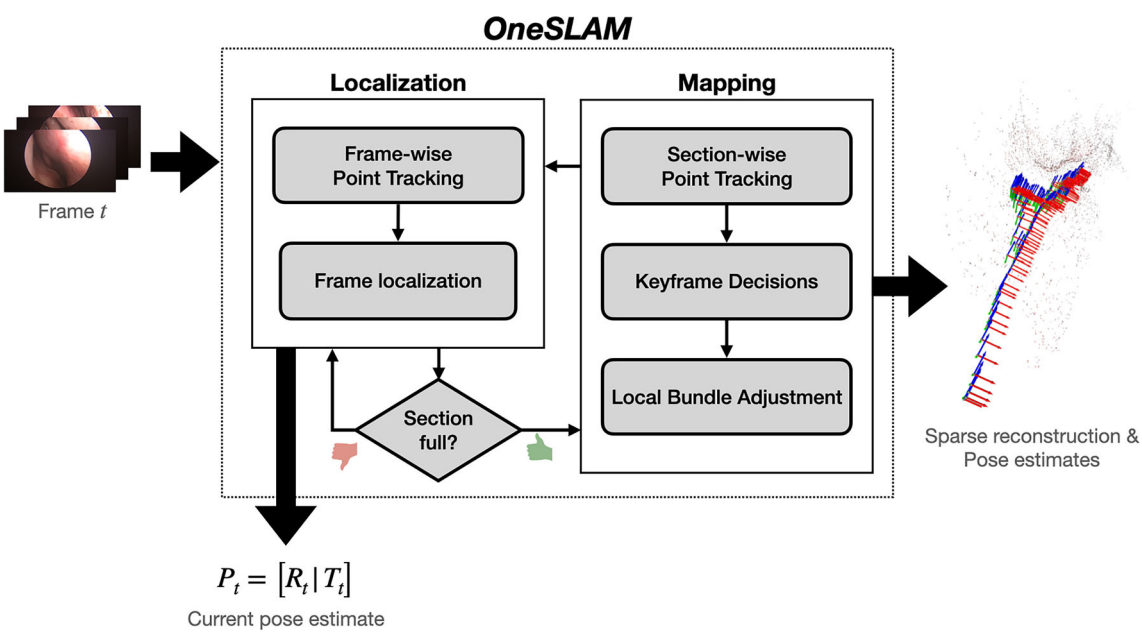

Fig. 1 Architecture of OneSLAM. We track keypoints for every new frame via zero-shot point tracking. To localize, we track a small set of keypoints onto the new frame and perform PnP pose estimation. In the mapping process, the key point tracks are refined by tracking across longer sections of the input video and used in a bundle adjustment optimization to obtain the keyframe poses and a sparse map of the anatomy

$$\sum_{\boldsymbol{x} \in K_T, X \in M} \|\boldsymbol{x} - \pi(P_T X)\| \tag{1}$$

where K_T is the set of key points on frame I_T, X is the 3D point associated with the key point $\boldsymbol{x}$, and π is the perspective projection to convert 3D world to pixel coordinates.

Mapping

After a total of $l_{\boldsymbol{m}}$ have been passed since the last mapping process, we start a new mapping process. First, this procedure improves the existing key point tracks for the frames in the section $V' = (I_{T-l_{\boldsymbol{m}}}, ..., I_T)$ by re-tracking the key point set $K_{T-l_{\boldsymbol{m}}}$ of frame $I_{T-l_{\boldsymbol{m}}}$ for all frames in V', providing Co-Tracker with the entire section V' to find the new tracks. In addition, a new set of key points $K^{\text{new}}_{T-l_{\boldsymbol{m}}}$ is generated and tracked as well, where the number of points $K^{\text{new}}_{T-l_{\boldsymbol{m}}}$ is chosen such that the total number of tracked points $|K^{\text{new}}_{T-l_{\boldsymbol{m}}}| + |K_{T-l_{\boldsymbol{m}}}|$ equals $n_{\boldsymbol{m}}$.

After generating refined key point tracks for the current section V', new keyframes are chosen from the section V', where we use simple keyframe subsample factor to decide if a frame should become a keyframe or not. Lastly, the mapping process starts optimizing the map M and keyframe poses $\boldsymbol{P} \subseteq \{P_t | t <= T\}$ by running bundle adjustment to minimize the reprojection error across frames. For our implementation, we employ `g2opy`, a python wrapper of the popular graph optimization engine `g2o` [28]. The optimization is performed in a sparse manner by running a Levenberg–Marquardt algorithm, with key frame poses that lie far back in time being fixated during the optimization.

Experimental setup

Datasets

We evaluate our approach on three different endoscopic dataset including sinus endoscopy, colonoscopy and arthroscopy. For **Sinus Endoscopy**, we use the SAGE-SLAM test dataset provided by [11], which is composed by three sinus endoscopic sequences. Additionally, we evaluate on an in-house sinus dataset [29], containing nine endoscopic sequences extracted from seven ex vivo subjects. In addition to the endoscopic sequences, the in-house dataset contains optical tracking information, and CT imaging, allowing for extensive comparison. For **Arthroscopy**, we use the Dense-ArthroSLAM dataset provided by Marmol et al. [14]. This dataset contains three arthroscopy sequences obtained from ex vivo subjects. Since both *OneSLAM* and EndoSLAM do not have any loop closure capacity, we decide to manually cut out the a local section for each video clip. Finally, for **Colonoscopy** we employ the C3VD dataset by Bobrow et al. [30] comprising of 22 colonoscopy sequences. The data, collected from phantom screening colonoscopy, contain registered ground truth information such as depth and camera poses. To evaluate the SLAM pipelines, we sample ten sequences from the available data. **Abdomen** We utilize the MITI dataset published by Hartwig et al. [33]. As with arthroscopy, we select local sections from the available sequences to account for the lack of loop closure in OneSLAM and EndoSLAM.

Tracking accuracy

We compare our proposed *OneSLAM* against three state-of-the-art domain-specific SLAM models, including the classical state-of-the-art algorithm ORB-SLAM3 [7], the EndoSLAM's motion estimation network EndoSFMLearner [15] and the sinus domain-specific method SAGE-SLAM [11]. To test the zero-shot attributes of our method, we employ the four endoscope domain dataset: the SAGE-SLAM test dataset for sinus endoscopy, Dense-ArthroSLAM dataset for arthroscopy and C3VD for the colon. Note that our approach does not require any training on endoscopic data.

For the learning-based model EndoSFMLearner, we maintained the default parameters and ensure a consistent hyper-parameter set when training across all four domains. To provide an upper bound to the performance of EndoSFMLearner, we train one model for each domain by selecting a representative set of the available sequences for training and validation. To obtain the final trajectory, we follow EndoSLAMs approach of concatenating all poses. For ORB-SLAM3, we iteratively fine-tune ORB parameters on a chosen representative sequence and subsequently apply the optimized ORB parameters to sequences within that domain. This process is repeated across all domains, and additional details can be found in Online Resource 1— Appendix C. In the sinus, we additionally evaluate our method against the pre-trained version of SAGE-SLAM provided by [11].

To compare the results, we compute the Absolute Trajectory Error (ATE) and the Relative Pose Error (RPE) as the average drift per mm. Note that since metric scale cannot be recovered from monocular images alone, the estimated trajectories are first scaled and aligned to groundtruth using the Umeyama algorithm [31]. We also provide the percentage of poses in the trajectory for which the approaches produced an estimate.

Further details regarding the implementation of the approaches can be found in Online Resource 1—Appendix C, E and F.

Sinus reconstruction accuracy

Additionally, we evaluate the reconstruction accuracy of the sparse maps generated by *OneSLAM*. To this end, we follow the process described in [29] to obtain the target registration error (TRE) and point-to-mesh distance (P2M) of our point clouds against segmented CT from the in-house dataset. To compute the registration between the CT and the point cloud, we perform a 50/50 split, where the first group of points was employed to register the geometries and the second group to compute the metrics.

Ablation study

Finally, we perform ablation studies to investigate the contributions of various strategies to the overall performance of our proposed pipeline. Specifically, we focus on components of the cross-frame point correspondences as they play pivotal roles in determining the mapping, which, in turn, directly affects the quality of motion recovery and reconstruction. For comparison, we provide the effective FPS, ATE and RPE between consecutive keyframes. Specifically, we ablate:

Sampling Strategy We evaluate five point sampling strategies. First, *UNIFORM* Samples points uniformly across the image. *DENSITY* assigns a higher likelihood of resampling to regions with low density of point tracks. Regions of low density are selected based on the aggregation of nine grids with sizes varying from two to nine. We also evaluate feature-based sampling strategies that consider the geometric information within image frames during the sampling phase. In this group, we evaluate the feature detectors *SIFT* [32], *ORB* [20] and *R2D2* [22], which rely on the criteria of distinctiveness and repeatability.

Number of tracked points We vary the number of tracked points n_m during mapping between 50, 200 and 800 to assess its influence on the performance. For sampling the points, we follow the *R2D2* strategy.

Results and discussion

Results in tracking accuracy show cross-domain generalizability

We first compare our performance against state-of-the-art methods that excel in their respective surgical domains. In qualitative terms, our proposed method *OneSLAM* shows comparable or improved performance in motion estimation compared with baseline SLAM methods as is presented in Table 1. Critically, our approach demonstrates zero-shot generalization to all of our tested domains, without sacrificing performance.

EndoSFMLearner is trained to optimize motion estimates between frames. As such, it only constrains the motion between adjacent frames I_t and I_{t+1}, leading the approach to quickly accumulate drift and loose track of the trajectory on a global scale, even when it is overfitted on the test data.

Table 1 Average tracking accuracy of our *OneSLAM* pipeline against state-of-the-art approaches to SLAM in endoscopy

Anatomy	Method	Trans. ATE [mm[1]]	Rot. ATE [deg]	Trans. RPE [mm[1]]	Rot. RPE [deg]	Cameras Tracked
Sinus	Ours	**1.20**	9.89	**0.29**	0.86	100%
	SAGE-SLAM	1.26	**4.54***	0.44	**0.84**	100%[2]
	EndoSLAM	4.72***	129.62***	2.46	2.11	100%[2]
	ORB-SLAM3	1.52	50.33**	0.55	2.28	75%
Colon	Ours	**0.63**	**11.96**	**0.29**	**0.37**	100%
	EndoSLAM	4.54***	115.85***	2.33	3.13	100%[2]
	ORB-SLAM3	1.74**	55.63***	1.10***	1.05**	74%
Joint	Ours	0.93	**16.07**	**0.49**	**1.05**	100%
	EndoSLAM	2.50	73.64*	1.09	14.38	100%[2]
	ORB-SLAM3	**0.80**	63.39	0.91*	1.67	20%
Abdomen	Ours	**3.58**	**19.55**	**1.21**	**0.40**	100%
	EndoSLAM	6.91	62.62*	5.94	4.02	100%[2]
	ORB-SLAM3	Fail	Fail	Fail	Fail	0%

Bold denotes the best observed value

[1] The sinus dataset was approximately scaled to mm using the average speed of the endoscope as reference. Colon, joint and abdomen are in mm

[2] Note that by design, SAGE-SLAM and EndoSLAM always return a pose estimate for every camera

In addition, we provide the percentage of poses that each method is able to produce an estimate for. *, ** and *** denote statistical significance compared to *OneSLAM* for $p < 1\%$, $p < 5\%$ and $p < 10\%$. All P-values are provided in Online Resource 1—Appendix G

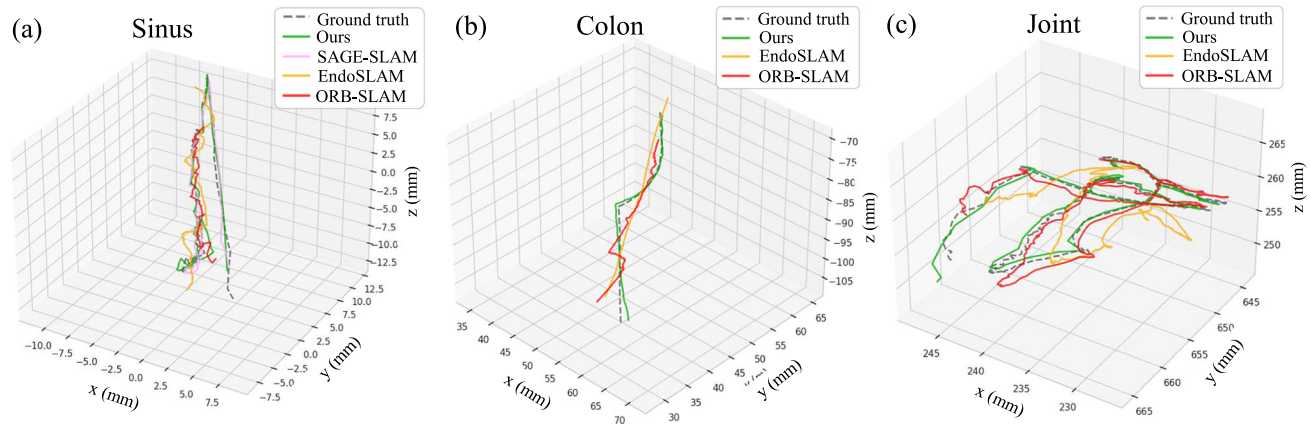

Fig. 2 Motion estimation results on data sequences in three surgical domains. **a** depicts trajectories estimated with OneSLAM, SAGE-SLAM, EndoSLAM and ORB-SLAM3 within one of the sinus data sequences. **b** depicts trajectories estimated with OneSLAM, EndoSLAM and ORB-SLAM3 on colon data, ORB-SLAM3 fails in this case. **c** showcases trajectories estimated with OneSLAM, EndoSLAM and ORB-SLAM3 on joint data

Table 2 Average target registration error (TRE) and point-to-mesh distance between the sparse reconstruction and the CT segmentation.

Method	Trans. ATE [mm]	Rot. ATE [deg]	TRE [mm]	Point-to-Mesh [mm]
Liu et al. [29]	2.20*	4.95	6.40	1.00
OneSLAM	3.52	5.65	6.22	1.26

Averages from a state-of-the-art SFM-based dense reconstruction pipeline by Liu et al. [33] are provided as reference. *, ** and *** denote statistical significance compared to *OneSLAM* for $p < 1\%$, $p < 5\%$ and $p < 10\%$. All P-values are provided in Online Resource 1— Appendix G

ORB-SLAM3's performance is notably inconsistent cross-domain. Moreover, we observe that it fails to consistently track the camera. We attribute these results to difficulties in detecting robust features on feature-scarce tissue surfaces. Consequently, it struggles to establish a strong initial state for tracking and mapping.

On the contrary, basing our approach on TAP foundation models allows us to ensure a substantial number of point correspondences persist across extensive frame sequences. This consistent point tracking yields advantages in pairwise frame alignment, while preventing an excessive presence of noisy 3D points that might otherwise cause local bundle adjustment to converge toward incorrect solutions. This is visually supported by Fig. 2 which shows the trajectory aligned with ground truth data for multiple domains.

Sparse reconstruction shows comparable results with SfM approaches

In Table 2, we present the average tracking and reconstruction accuracy on our in-house sinus dataset. The table presents the P2M distance and target registration error obtained with *OneSLAM*, compared with the SfM-based reconstruction discussed in [29]. Results presented in Table 2 suggest that our pipeline manages to produce competitive results, but overall does not yet meet the often-cited clinical requirement of sub-millimeter tracking and reconstruction accuracy [34]. We attribute this error mainly to the accumulation of drift, as our current pipeline lacks a loop closure mechanism to ensure global consistency. Consequently, future work will be focused on extending our pipeline to be able to detect and close loops.

Table 3 Ablation over the number of tracked points n_m for the colon

n_m	FPS	Trans. ATE [mm]	Rot. ATE [deg]	Trans. RPE [mm]	Rot. RPE [deg]
50	**2.13**	0.93	62.55	0.29	0.35
200	1.93	0.63	**11.96**	0.15	0.16
800	1.59	**0.60**	13.26	**0.14**	**0.13**

Bold denotes the best observed value

Ablation study

Number of tracked points Table 3 shows the results for our ablation study regarding the number of tracked points. Generally, we observe that a larger number of points lead to more accurate trajectories, but with diminishing returns as the number of points increases. At the same time, since CoTracker scales quadratically in the number of tracked points [25], we observe significant decreases in FPS with increasing point number.

Point sampling strategy As seen in Table 4, both the *SIFT* and *ORB* sampling strategies perform significantly worse com-

Table 4 Ablation over the sampling strategy for the colon

Strategy	FPS	Trans. ATE [mm]	Rot. ATE [deg]	Trans. RPE [mm]	Rot. RPE [deg]
UNIFORM	**2.04**	0.74	15.39	0.20	0.26
DENSITY	2.01	0.67	18.75	**0.15**	**0.16**
R2D2	1.93	**0.63**	**11.96**	**0.15**	**0.16**
SIFT	1.97	1.71	79.39	0.51	0.84
ORB	2.03	2.91	80.51	0.68	1.02

Bold denotes the best observed value

pared to the other sampling strategies, due to the samples being clustered around significant image features such as specular highlights, which in turn reduces the tracking accuracy. In the colon, between *DENSITY* and *UNIFORM*, the former manages to outperform the latter in three out of four metrics, indicating that *DENSITY* performs more robustly than *UNIFORM*. Overall, the *R2D2* strategy performs best, due to being able to sample large number of meaningful image features. We do not observe a significant impact on FPS coming from the sampling strategy, as new points are only sampled in every mapping iteration.

Limitations and failure cases

Limitations The primary limitation of *OneSLAM* is the lack of a loop closure mechanism, which results in strong drift accumulation as the pipeline processes longer sequences. Additionally, we note that our pipeline is not yet real-time capable, as it is currently running at an average speed of around two to four effective FPS. Notably, the localization and mapping procedure run at significantly different speeds, in turn causing significant variance in the framerate. Finally, our approach currently assumes the underlying scene to be static. Therefore, it is limited to scenes without deformation or tool movement. We note that the evaluation of *OneSLAM* is constrained to the video modalities we have used in our evaluation. Videos from different sensors [35] may exhibit distinct characteristics, potentially leading to variations in performance.

Failure Cases Notably, *OneSLAM* depends on the output of the TAP foundation model to optimize camera poses. As such, it fails when the TAP foundation model fails, such as in the case of very blurry images or very fast camera movement that results in strong jumps between individual frames. Moreover, partly as a result of the missing loop closure mechanism, *OneSLAM* performs significantly worse if keypoints are lost to quickly. This is most notable when combining fast camera motion with a low field of view, as here *OneSLAM* might not resample new keypoints fast enough to make up for the ones lost, causing the pipeline to fail completely. Due to the same reason, pivoting motion is also more challenging for *OneSLAM* than backwards and forwards motion, as in the latter case keypoints tend to stay in view for longer.

Conclusion

We present *OneSLAM*, a monocular SLAM pipeline for endoscopy leveraging zero-shot point tracking to perform camera tracking and 3D scene reconstruction in various surgical domains. Our results show that our pipeline out of the box generalizes cross-domain without any additional fine-tuning and without sacrificing performance.

However, while our method produces promising results, it is limited by a lack of a global loop detection algorithm, causing it to accumulate drift when running on longer sequences. In addition, the pipeline is not yet capable of performing in real time, running at around two to four FPS with significant variance in the framerate. To address these issues, future work will focus on incorporating the methodology of Moing et al. [36] to reduce the number of total point tracks, updating our pipeline to take advantage of the recently published online point tracking capabilities of CoTracker and the development of a loop closure mechanism to ensure global consistency.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11548-024-03171-6.

Funding This work was funded in part by Johns Hopkins University internal funds and in part by NIH R01EB030511. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethical approval Not required for this work.

Code availability The source code is available at https://github.com/arcadelab/OneSLAM.

Consent to participate This study was performed under the approved IRB00267324 protocol on non-living subjects, for which informed consent was not required.

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